CHAPTER 5

VIRTUAL MACHINES PROVISIONING AND MIGRATION SERVICES

MOHAMED EL-REFAEY

5.1 INTRODUCTION AND INSPIRATION

Cloud computing is an emerging research infrastructure that builds on the achievements of different research areas, such as service-oriented architecture (SOA), grid computing, and virtualization technology. It offers infrastructure as a service that is based on pay-as-you-use and on-demand computing models to the end users (exactly the same as a public utility service like electricity, water, gas, etc.). This service is referred to as Infrastructure as a Service (IaaS). To provide this cloud computing service, the provisioning of the cloud infrastructure in data centers is a prerequisite. However, the provisioning for systems and applications on a large number of physical machines is traditionally a time-consuming process with low assurance on deployment's time and cost.

In this chapter, we shall focus on two core services that enable the users to get the best out of the IaaS model in public and private cloud setups. These services are named virtual machine provisioning and migration services. We will also cover their concepts, techniques, and research directions, along with an introductory overview about virtualization technology and its role as a fundamental component/block of the cloud computing architecture stack.

To make the concept clearer, consider this analogy for virtual machine provisioning, to know its value: *Historically*, when there is a need to install a new server for a certain workload to provide a particular service for a client, lots of effort was exerted by the IT administrator, and much time was spent to install and provision a new server, because the administrator has to follow specific checklist and procedures to perform this task on hand. (Check the

Cloud Computing: Principles and Paradigms, Edited by Rajkumar Buyya, James Broberg and Andrzej Goscinski

inventory for a new machine, get one, format, install OS required, and install services; a server is needed along with lots of security batches and appliances.) **Now,** with the emergence of virtualization technology and the cloud computing IaaS model, it is just a matter of minutes to achieve the same task. All you need is to provision a virtual server through a self-service interface with small steps to get what you desire with the required specifications—whether you are provisioning this machine in a public cloud like Amazon Elastic Compute Cloud (EC2) or using a virtualization management software package or a private cloud management solution installed at your data center in order to provision the virtual machine inside the organization and within the private cloud setup. This scenario is an awesome example for illustrating the value of virtualization and the way virtual machines are provisioned.

We can draw the same analogy for migration services. *Previously*, whenever there was a need for performing a server's upgrade or performing maintenance tasks, you would exert a lot of time and effort, because it is an expensive operation to maintain or upgrade a main server that has lots of applications and users. Now, with the advance of the revolutionized virtualization technology and migration services associated with hypervisors' capabilities, these tasks (maintenance, upgrades, patches, etc.) are very easy and need no time to accomplish.

Provisioning a new virtual machine is a matter of minutes, saving lots of time and effort. Migrations of a virtual machine is a matter of milliseconds: saving time, effort, making the service alive for customers, and achieving the SLA/SLO agreements and quality-of-service (QoS) specifications required.

An overview about the chapter's higlights and sections can be grasped by the mind map shown in Figure 5.1.

5.2 BACKGROUND AND RELATED WORK

In this section, we will have a quick look at previous work, give an overview about virtualization technology, public cloud, private cloud, standardization efforts, high availability through the migration, and provisioning of virtual machines, and shed some lights on distributed management's tools.

5.2.1 Virtualization Technology Overview

Virtualization has revolutionized data center's technology through a set of techniques and tools that facilitate the providing and management of the dynamic data center's infrastructure. It has become an essential and enabling technology of cloud computing environments. Virtualization can be defined as the abstraction of the four computing resources (storage, processing power, memory, and network or I/O). It is conceptually similar to emulation, where a system pretends to be another system, whereas virtualization is a system pretending to be two or more of the same system [1]. As shown in

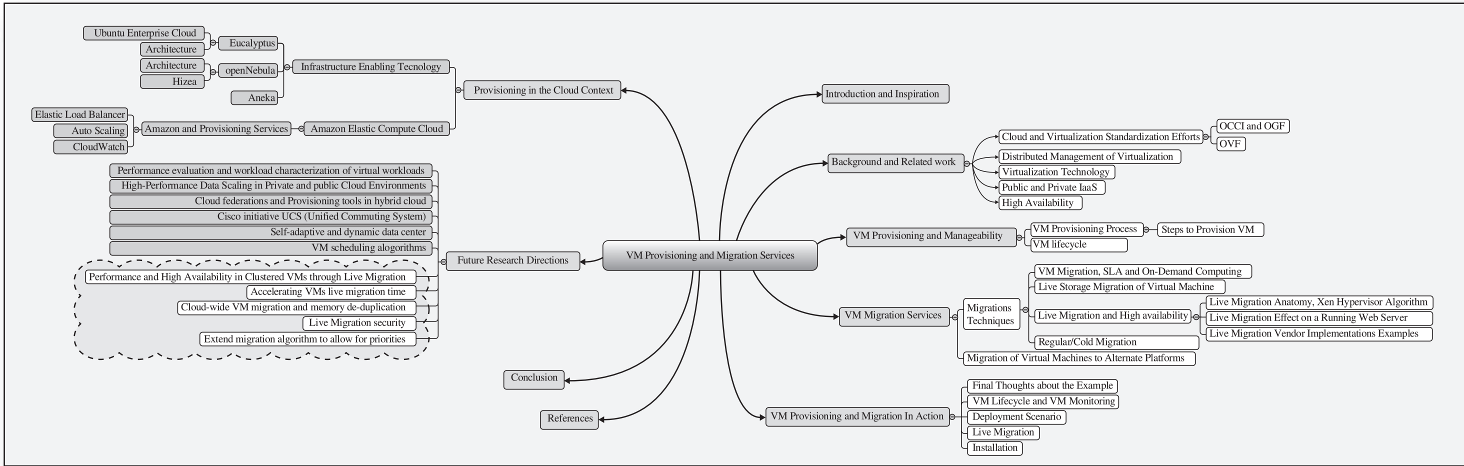

FIGURE 5.1. VM provisioning and migration mind map.

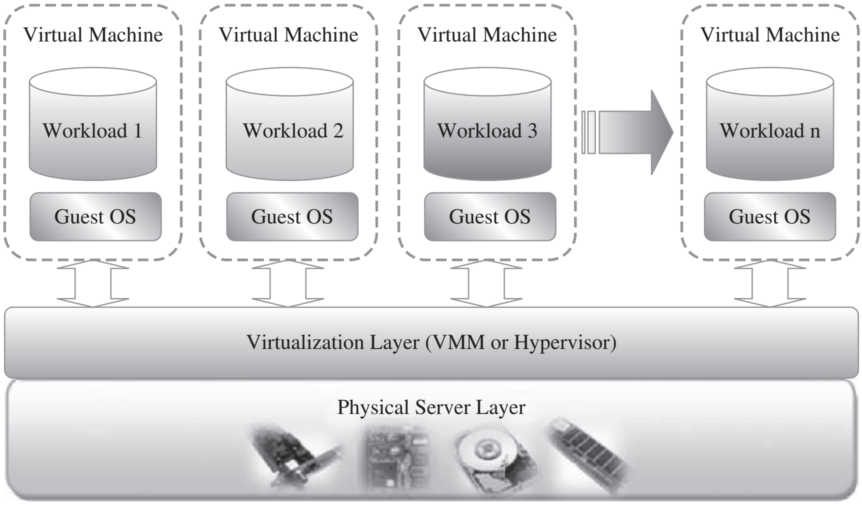

FIGURE 5.2. A layered virtualization technology architecture.

Figure 5.2, the virtualization layer will partition the physical resource of the underlying physical server into multiple virtual machines with different workloads. The fascinating thing about this virtualization layer is that it schedules, allocates the physical resource, and makes each virtual machine think that it totally owns the whole underlying hardware's physical resource (processor, disks, rams, etc.)[2].

Virtual machine's technology makes it very flexible and easy to manage resources in cloud computing environments, because they improve the utilization of such resources by multiplexing many virtual machines on one physical host (server consolidation), as shown in Figure 5.1. These machines can be scaled up and down on demand with a high level of resources' abstraction.

Virtualization enables high, reliable, and agile deployment mechanisms and management of services, providing on-demand cloning and live migration services which improve reliability. Accordingly, having an effective management's suite for managing virtual machines' infrastructure is critical for any cloud computing infrastructure as a service (IaaS) vendor.

5.2.2 Public Cloud and Infrastructure Services

Public cloud or external cloud describes cloud computing in a traditional mainstream sense, whereby resources are dynamically provisioned via publicly accessible Web applications/Web services (SOAP or RESTful interfaces) from an off-site third-party provider, who shares resources and bills on a fine-grained utility computing basis [3], the user pays only for the capacity of the provisioned resources at a particular time.

There are many examples for vendors who publicly provide infrastructure as a service. Amazon Elastic Compute Cloud (EC2)[4] is the best known example, but the market now bristles with lots of competition like GoGrid [5], Joyent Accelerator [6], Rackspace [7], AppNexus [8], FlexiScale [9], and Manjrasoft Aneka [10].

Here, we will briefly cover and describe Amazon EC2 offering. Amazon Elastic Compute Cloud (EC2) is an IaaS service that provides elastic compute capacity in the cloud. These services can be leveraged via Web services (SOAP or REST), a Web-based AWS (Amazon Web Service) management console, or the EC2 command line tools. The Amazon service provides hundreds of pre-made AMIs (Amazon Machine Images) with a variety of operating systems (i.e., Linux, OpenSolaris, or Windows) and pre-loaded software.

It provides you with complete control of your computing resources and lets you run on Amazon's computing and infrastructure environment easily. Amazon EC2 reduces the time required for obtaining and booting a new server's instances to minutes, thereby allowing a quick scalable capacity and resources, up and down, as the computing requirements change. Amazon offers different instances' size according to (a) the resources' needs (small, large, and extra large), (b) the high CPU's needs it provides (medium and extra large high CPU instances), and (c) high-memory instances (extra large, double extra large, and quadruple extra large instance).

5.2.3 Private Cloud and Infrastructure Services

A private cloud aims at providing public cloud functionality, but on private resources, while maintaining control over an organization's data and resources to meet security and governance's requirements in an organization. Private cloud exhibits a highly virtualized cloud data center located inside your organization's firewall. It may also be a private space dedicated for your company within a cloud vendor's data center designed to handle the organization's workloads.

Private clouds exhibit the following characteristics:

- Allow service provisioning and compute capability for an organization's users in a self-service manner.
- Automate and provide well-managed virtualized environments.
- Optimize computing resources, and servers' utilization.
- Support specific workloads.

There are many examples for vendors and frameworks that provide infrastructure as a service in private setups. The best-known examples are Eucalyptus [11] and OpenNebula [12] (which will be covered in more detail later on).

It is also important to highlight a third type of cloud setup named "hybrid cloud," in which a combination of private/internal and external cloud resources exist together by enabling outsourcing of noncritical services and functions in public cloud and keeping the critical ones internal. Hybrid cloud's main function is to release resources from a public cloud and to handle sudden demand usage, which is called "cloud bursting."

5.2.4 Distributed Management of Virtualization

Virtualization's benefits bring their own challenges and complexities presented in the need for a powerful management capabilities. That is why many commercial, open source products and research projects such as OpenNebula [12], IBM Virtualization Manager, Joyent, and VMware DRS are being developed to dynamically provision virtual machines, utilizing the physical infrastrcture. There are also some commercial and scientific infrastructure cloud computing initiatives, such as Globus VWS, Eucalyptus [11] and Amazon, which provide remote interfaces for controling and monitoring virtual resources. One more effort in this context is the RESERVOIR [13] initiative, in which grid interfaces and protocols enable the required interoperability between the clouds or infrastructure's providers. RESERVOIR also, needs to expand substantially on the current state-of-the-art for grid-wide accounting [14], and to increase the flexibility of supporting different billing schemes, and accounting for services with indefinite lifetime, as opposed to finite jobs with support to account for utilization metrics relevant to virtual machines [15].

5.2.5 High Availability

High availability is a system design protocol and an associated implementation that ensures a certain absolute degree of operational continuity during a given measurement period. Availability refers to the ability of a user's community to access the system—whether for submiting new work, updating or altering existing work, or collecting the results of the previous work. If a user cannot access the system, it is said to be unavailable [16]. This means that services should be available all the time along with some planned/unplanned downtime according to a certain SLA (formalize the service availaibiliy objectives, and requirments) which often refers to the monthly availability or downtime of a service; to calculate the service's credits to match the billing cycles. Services that are considered as business critical are often categorized as *high availability* services. Systems running business critical services should be planned and designed from the bottom with the goal of achieving the lowest possible amount of planned and unplanned downtime.

Since a virtual environment is the larger part of any organization, management of these virtual resources within this environemnet becomes a critical mission, and the migration services of these resources became a corner stone in achieving high availability for these services hosted by VMs. So, in the context of virtualized infrastructure, high availability allows virtual machines to automatically be restarted in case of an underlying hardware failure or individual VM failure. If one of your servers fails, the VMs will be restarted on other virtualized servers in the resource pool, restoring the essential services with minimal service interruption.

5.2.6 Cloud and Virtualization Standardization Efforts

Standardization is important to ensure interoperability between virtualization mangement vendors, the virtual machines produced by each one of them, and cloud computing. Here, we will have look at the prevalent standards that make cloud computing and virtualization possible. In the past few years, virtualization standardization efforts led by the Distributed Management Task Force (DMTF) have produced standards for almost all the aspects of virtualization technology. DMTF initiated the VMAN (Virtualization Management Initiative), which delivers broadly supported interoperability and portability standards for managing the virtual computing lifecycle. VMAN's OVF (Open Virtualization Format) in a collaboration between industry key players: Dell, HP, IBM, Microsoft, XenSource, and Vmware. OVF specification provides a common format to package and securely distribute virtual appliances across multiple virtualization platforms. VMAN profiles define a consistent way of managing a heterogeneous virtualized environment [17].

5.2.7 OCCI and OGF

Another standardization effort has been initiated by Open Grid Forum (OGF) through organizing an official new working group to deliver a standard API for cloud IaaS, the Open Cloud Computing Interface Working Group (OCCI-WG). This group is dedicated for delivering an API specification for the remote management of cloud computing's infrastructure and for allowing the development of interoperable tools for common tasks including deployment, autonomic scaling, and monitoring. The scope of the specification will be covering a high-level functionality required for managing the life-cycle virtual machines (or workloads), running on virtualization technologies (or containers), and supporting service elasticity. The new API for interfacing "IaaS" cloud computing facilities will allow [18]:

- **Consumers** to interact with cloud computing infrastructure on an ad hoc basis.
- **Integrators** to offer advanced management services.
- **Aggregators** to offer a single common interface to multiple providers.
- **Providers** to offer a standard interface that is compatible with the available tools.

- **Vendors** of grids/clouds to offer standard interfaces for dynamically scalable service's delivery in their products.

5.3 VIRTUAL MACHINES PROVISIONING AND MANAGEABILITY

In this section, we will have an overview on the typical life cycle of VM and its major possible states of operation, which make the management and automation of VMs in virtual and cloud environments easier than in traditional computing environments.

As shown in Figure 5.3, the cycle starts by a request delivered to the IT department, stating the requirement for creating a new server for a particular service. This request is being processed by the IT administration to start seeing the servers' resource pool, matching these resources with the requirements, and starting the provision of the needed virtual machine. Once it is provisioned and started, it is ready to provide the required service according to an SLA, or a time period after which the virtual is being released; and free resources, in this case, won't be needed.

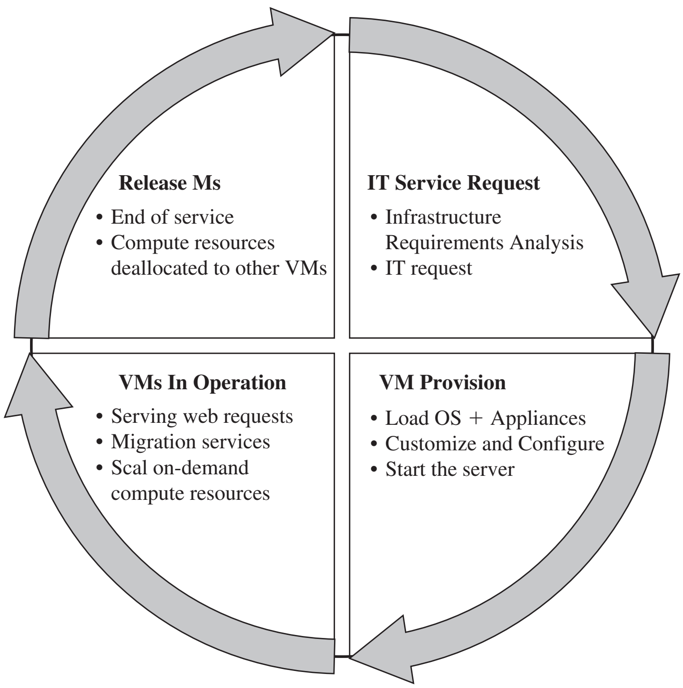

FIGURE 5.3. Virtual machine life cycle.

5.3.1 VM Provisioning Process

Provisioning a virtual machine or server can be explained and illustrated as in Figure 5.4:

Steps to Provision VM. Here, we describe the common and normal steps of provisioning a virtual server:

- Firstly, you need to select a server from a pool of available servers (physical servers with enough capacity) along with the appropriate OS template you need to provision the virtual machine.
- Secondly, you need to load the appropriate software (operating system you selected in the previous step, device drivers, middleware, and the needed applications for the service required).
- Thirdly, you need to customize and configure the machine (e.g., IP address, Gateway) to configure an associated network and storage resources.
- Finally, the virtual server is ready to start with its newly loaded software.

Typically, these are the tasks required or being performed by an IT or a data center's specialist to provision a particular virtual machine.

To summarize, server provisioning is defining server's configuration based on the organization requirements, a hardware, and software component (processor, RAM, storage, networking, operating system, applications, etc.). Normally, virtual machines can be provisioned by manually installing an operating system, by using a preconfigured VM template, by cloning an existing VM, or by importing a physical server or a virtual server from another hosting platform. Physical servers can also be virtualized and provisioned using P2V (physical to virtual) tools and techniques (e.g., virt-p2v).

After creating a virtual machine by virtualizing a physical server, or by building a new virtual server in the virtual environment, a template can be created out of it. Most virtualization management vendors (VMware, XenServer, etc.) provide the data center's administration with the ability to do such tasks in

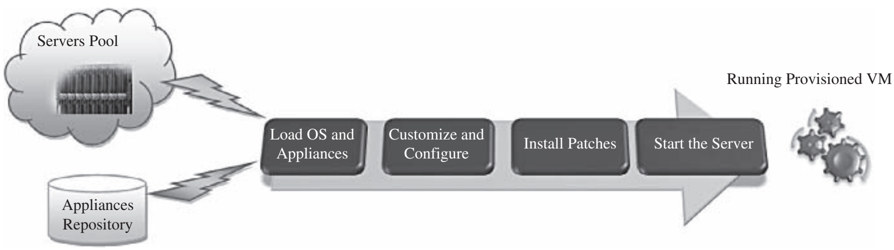

FIGURE 5.4. Virtual machine provision process.

an easy way. Provisioning from a template is an invaluable feature, because it reduces the time required to create a new virtual machine.

Administrators can create different templates for different purposes. For example, you can create a Windows 2003 Server template for the finance department, or a Red Hat Linux template for the engineering department. This enables the administrator to quickly provision a correctly configured virtual server on demand.

This ease and flexibility bring with them the problem of virtual machine's sprawl, where virtual machines are provisioned so rapidly that documenting and managing the virtual machine's life cycle become a challenge [9].

5.4 VIRTUAL MACHINE MIGRATION SERVICES

Migration service, in the context of virtual machines, is the process of moving a virtual machine from one host server or storage location to another; there are different techniques of VM migration, hot/life migration, cold/regular migration, and live storage migration of a virtual machine [20]. In this process, all key machines' components, such as CPU, storage disks, networking, and memory, are completely virtualized, thereby facilitating the entire state of a virtual machine to be captured by a set of easily moved data files. We will cover some of the migration's techniques that most virtualization tools provide as a feature.

5.4.1 Migrations Techniques

Live Migration and High Availability. *Live migration* (which is also called hot or real-time migration) can be defined as the movement of a virtual machine from one physical host to another while being powered on. When it is properly carried out, this process takes place without any noticeable effect from the end user's point of view (a matter of milliseconds). One of the most significant advantages of live migration is the fact that it facilitates proactive maintenance in case of failure, because the potential problem can be resolved before the disruption of service occurs. Live migration can also be used for load balancing in which work is shared among computers in order to optimize the utilization of available CPU resources.

Live Migration Anatomy, Xen Hypervisor Algorithm. In this section we will explain live migration's mechanism and how memory and virtual machine states are being transferred, through the network, from one host A to another host B [21]; the Xen hypervisor is an example for this mechanism. The logical steps that are executed when migrating an OS are summarized in Figure 5.5. In this research, the migration process has been viewed as a transactional interaction between the two hosts involved:

Stage 0: Pre-Migration. An active virtual machine exists on the physical host A.

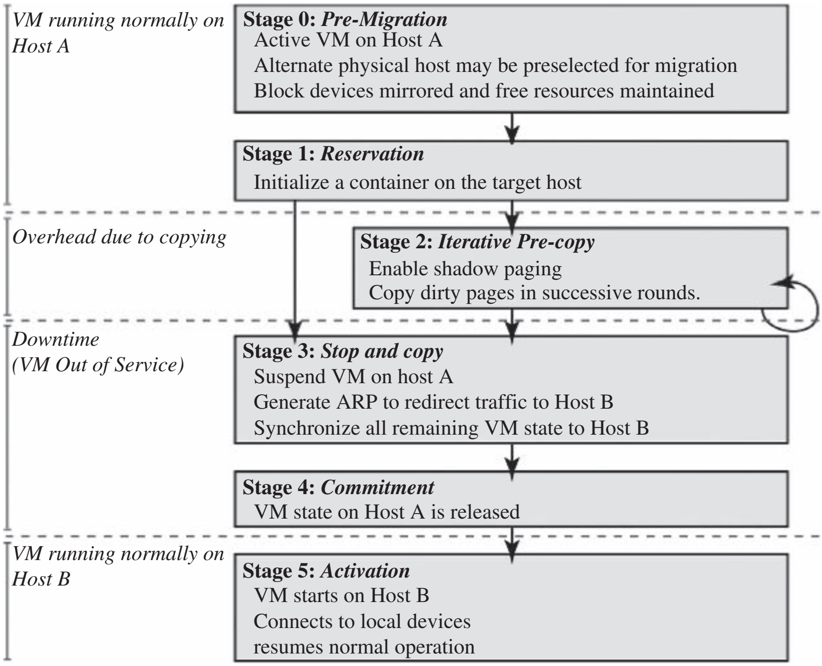

FIGURE 5.5. Live migration timeline [21].

Stage 1: Reservation. A request is issued to migrate an OS from host A to host B (a precondition is that the necessary resources exist on B and on a VM container of that size).

Stage 2: Iterative Pre-Copy. During the first iteration, all pages are transferred from A to B. Subsequent iterations copy only those pages dirtied during the previous transfer phase.

Stage 3: Stop-and-Copy. Running OS instance at A is suspended, and its network traffic is redirected to B. As described in reference 21, CPU state and any remaining inconsistent memory pages are then transferred. At the end of this stage, there is a consistent suspended copy of the VM at both A and B. The copy at A is considered primary and is resumed in case of failure.

Stage 4: Commitment. Host B indicates to A that it has successfully received a consistent OS image. Host A acknowledges this message as a commitment of the migration transaction. Host A may now discard the original VM, and host B becomes the primary host.

Stage 5: Activation. The migrated VM on B is now activated. Post-migration code runs to reattach the device's drivers to the new machine and advertise moved IP addresses.

This approach to failure management ensures that at least one host has a consistent VM image at all times during migration. It depends on the assumption that the original host remains stable until the migration commits and that the VM may be suspended and resumed on that host with no risk of failure. Based on these assumptions, a migration request essentially attempts to move the VM to a new host and on any sort of failure, execution is resumed locally, aborting the migration.

Live Migration Effect on a Running Web Server. Clark et al. [21] did evaluate the above migration on an Apache 1.3 Web server; this served static content at a high rate, as illustrated in Figure 5.6. The throughput is achieved when continuously serving a single 512-kB file to a set of one hundred concurrent clients. The Web server virtual machine has a memory allocation of 800 MB. At the start of the trace, the server achieves a consistent throughput of approximately 870 Mbit/sec. Migration starts 27 sec into the trace, but is initially rate-limited to 100 Mbit/sec (12% CPU), resulting in server's throughput drop to 765 Mbit/sec. This initial low-rate pass transfers 776 MB and lasts for 62 sec. At this point, the migration's algorithm, described in Section 5.4.1, increases its rate over several iterations and finally suspends the VM after a further 9.8 sec. The final stop-and-copy phase then transfers the remaining pages, and the Web server resumes at full rate after a 165-msec outage.

This simple example demonstrates that a highly loaded server can be migrated with both controlled impact on live services and a short downtime. However, the working set of the server, in this case, is rather small. So, this should be expected as a relatively easy case of live migration.

Live Migration Vendor Implementations Examples. There are lots of VM management and provisioning tools that provide the live migration of VM facility, two of which are VMware VMotion and Citrix XenServer "XenMotion."

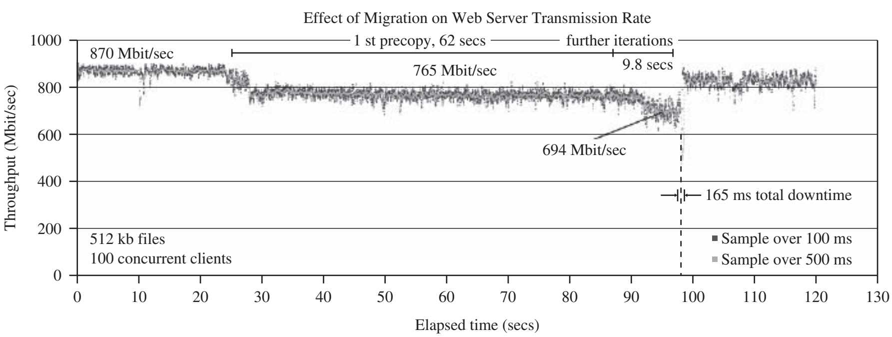

FIGURE 5.6. Results of migrating a running Web server VM [21].

VMware Vmotion. This allows users to (a) automatically optimize and allocate an entire pool of resources for maximum hardware utilization, flexibility, and availability and (b) perform hardware's maintenance without scheduled downtime along with migrating virtual machines away from failing or underperforming servers [22].

Citrix XenServer XenMotion. This is a nice feature of the Citrix XenServer product, inherited from the Xen live migrate utility, which provides the IT administrator with the facility to move a running VM from one XenServer to another in the same pool without interrupting the service (hypothetically for zero-downtime server maintenance, which actually takes minutes), making it a highly available service. This also can be a good feature to balance the workloads on the virtualized environment [23].

Regular/Cold Migration. Cold migration is the migration of a powered-off virtual machine. With cold migration, you have the option of moving the associated disks from one data store to another. The virtual machines are not required to be on a shared storage. It's important to highlight that the two main differences between live migration and cold migration are that live migration needs a shared storage for virtual machines in the server's pool, but cold migration does not; also, in live migration for a virtual machine between two hosts, there would be certain CPU compatibility checks to be applied; while in cold migration this checks do not apply. The cold migration process is simple to implement (as the case for the VMware product), and it can be summarized as follows [24]:

- The configuration files, including the NVRAM file (BIOS settings), log files, as well as the disks of the virtual machine, are moved from the source host to the destination host's associated storage area.
- The virtual machine is registered with the new host.
- After the migration is completed, the old version of the virtual machine is deleted from the source host.

Live Storage Migration of Virtual Machine. This kind of migration constitutes moving the virtual disks or configuration file of a running virtual machine to a new data store without any interruption in the availability of the virtual machine's service. For more details about how this option is working in a VMware product, see reference 20.

5.4.2 VM Migration, SLA and On-Demand Computing

As we discussed, virtual machines' migration plays an important role in data centers by making it easy to adjust resource's priorities to match resource's demand conditions.

This role is completely going in the direction of meeting SLAs; once it has been detected that a particular VM is consuming more than its fair share of resources at the expense of other VMs on the same host, it will be eligible, for this machine, to either be moved to another underutilized host or assign more resources for it, in case that the host machine still has resources; this in turn will highly avoid the violations of the SLA and will also, fulfill the requirements of on-demand computing resources. In order to achieve such goals, there should be an integration between virtualization's management tools (with its migrations and performance's monitoring capabilities), and SLA's management tools to achieve balance in resources by migrating and monitoring the workloads, and accordingly, meeting the SLA.

5.4.3 Migration of Virtual Machines to Alternate Platforms

One of the nicest advantages of having facility in data center's technologies is to have the ability to migrate virtual machines from one platform to another; there are a number of ways for achieving this, such as depending on the source and target virtualization's platforms and on the vendor's tools that manage this facility—for example, the VMware converter that handles migrations between ESX hosts; the VMware server; and the VMware workstation. The VMware converter can also import from other virtualization platforms, such as Microsoft virtual server machines [9].

5.5 VM PROVISIONING AND MIGRATION IN ACTION

Now, it is time to get into business with a real example of how we can manage the life cycle, provision, and migrate a virtual machine by the help of one of the open source frameworks used to manage virtualized infrastructure. Here, we will use ConVirt [25] (open source framework for the management of open source virtualization like Xen [26] and KVM [27], known previously as *XenMan*).

Deployment Scenario. ConVirt deployment consists of at least one ConVirt workstation, where ConVirt is installed and ran, which provides the main console for managing the VM life cycle, managing images, provisioning new VMs, monitoring machine resources, and so on. There are two essential deployment scenarios for ConVirt: A, *basic* configuration in which the Xen or KVM virtualization platform is on the local machine, where ConVirt is already installed; B, an *advanced* configuration in which the Xen or KVM is on one or more remote servers. The scenario in use here is the advanced one. In data centers, it is very common to install centralized management software (ConVirt here) on a dedicated machine for use in managing remote servers in the data center. In our example, we will use this dedicated machine where ConVirt is installed and used to manage a pool of remote servers (two machines). In order to use advanced features of ConVirt (e.g., live migration), you should set up a shared storage for the server pool in use on which the disks of the provisioned virtual machines are stored. Figure 5.7 illustrates the scenario.

Installation. The installation process involves the following:

- Installing ConVirt on at least one computer. See reference 28 for installation details.
- Preparing each managed server to be managed by ConVirt. See reference 28 for managed servers' installation details. We have two managing servers with the following Ips (managed server 1, IP:172.16.2.22; and managed server 2, IP:172.16.2.25) as shown in the deployment diagram (Figure 5.7).
- Starting ConVirt and discovering the managed servers you have prepared.

Notes

- Try to follow the installation steps existing in reference 28 according to the distribution of the operating system in use. In our experiment, we use Ubuntu 8.10 in our setup.
- Make sure that the managed servers include Xen or KVM hypervisors installed.
- Make sure that you can access managed servers from your ConVirt management console through SSH.

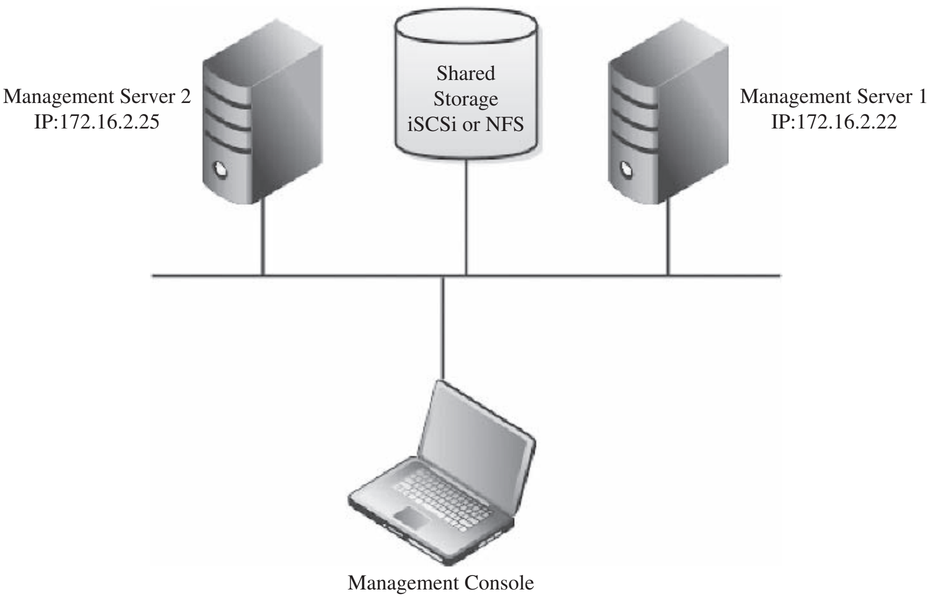

FIGURE 5.7. A deployment scenario network diagram.

Environment, Software, and Hardware. ConVirt 1.1, Linux Ubuntu 8.10, three machines, Dell core 2 due processor, 4G RAM.

Adding Managed Servers and Provisioning VM. Once the installation is done and you are ready to manage your virtual infrastructure, then you can start the ConVirt management console (see Figure 5.8):

Select any of servers' pools existing (QA Lab in our scenario) and on its context menu, select "Add Server."

- You will be faced with a message asking about the virtualization platform you want to manage (Xen or KVM), as shown in Figure 5.9:
- Choose KVM, and then enter the managed server information and credentials (IP, username, and password) as shown in Figure 5.10.
- Once the server is synchronized and authenticated with the management console, it will appear in the left pane/of the ConVirt, as shown in Figure 5.11.
- Select this server, and start provisioning your virtual machine as in Figure 5.12:
- Fill in the virtual machine's information (name, storage, OS template, etc.; Figure 5.13); then you will find it created on the managed server tree powered-off.

Note: While provisioning your virtual machine, make sure that you create disks on the shared storage (NFS or iSCSi). You can do so by selecting

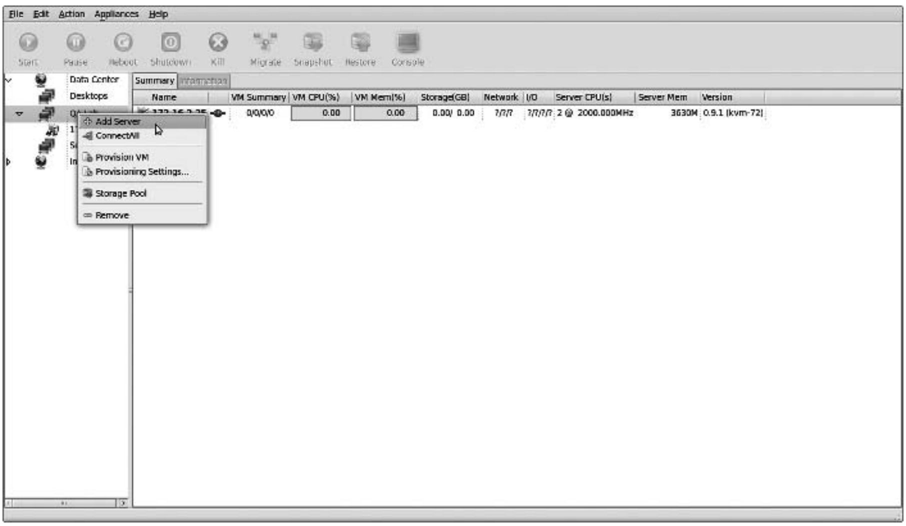

FIGURE 5.8. Adding managed server on the data centre's management console.

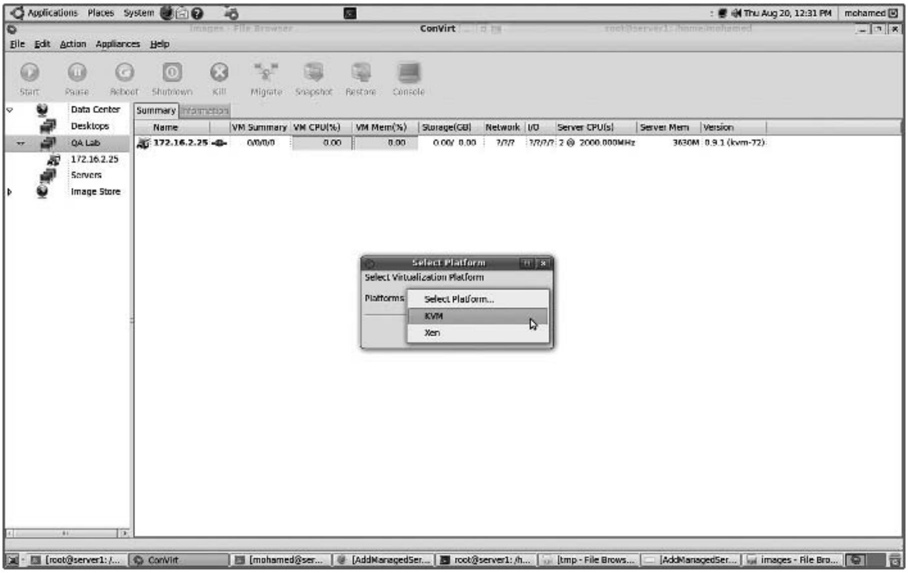

FIGURE 5.9. Select virtualization platform.

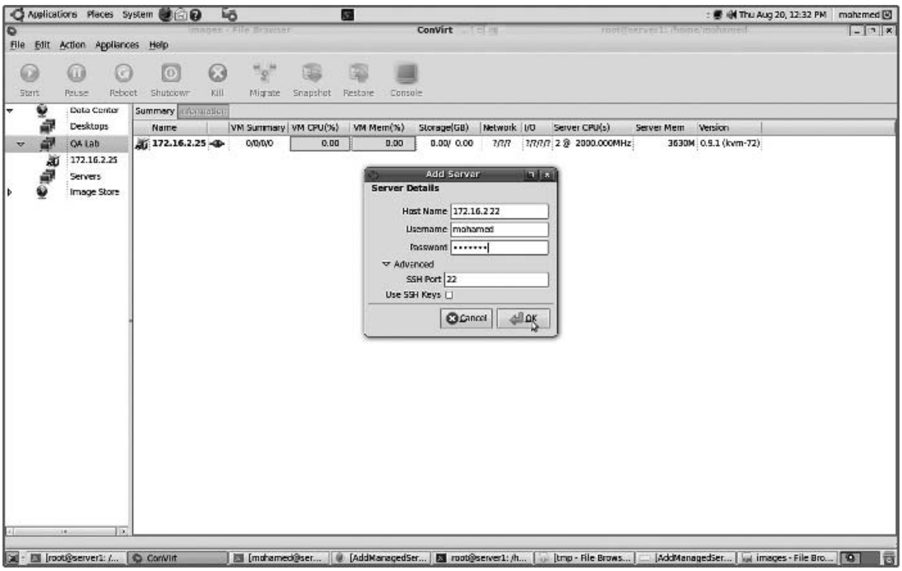

FIGURE 5.10. Managed server info and credentials.

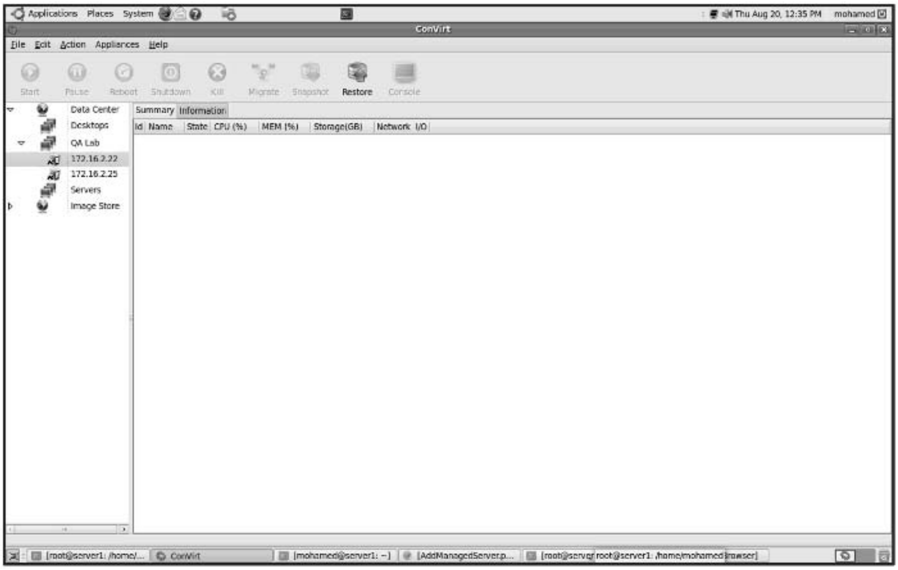

FIGURE 5.11. Managed server has been added.

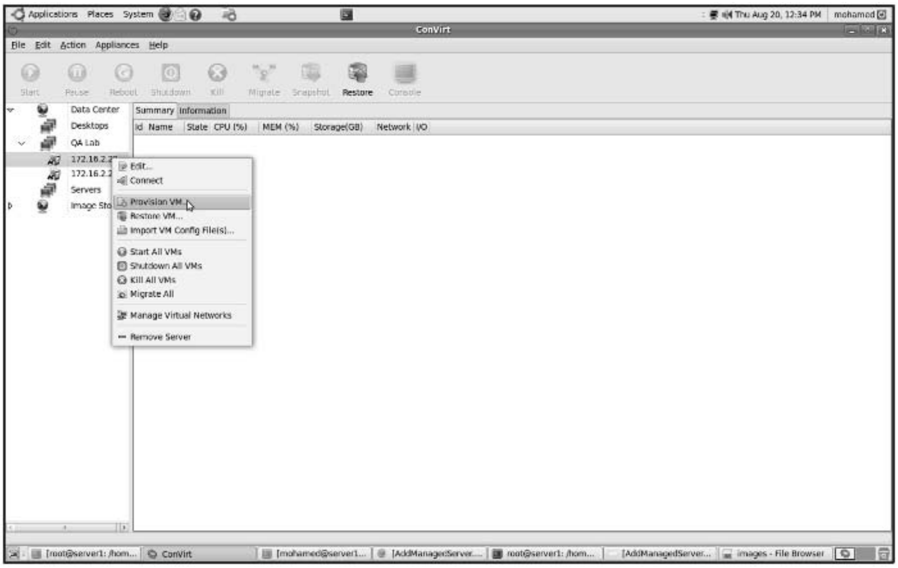

FIGURE 5.12. Provision a virtual machine.

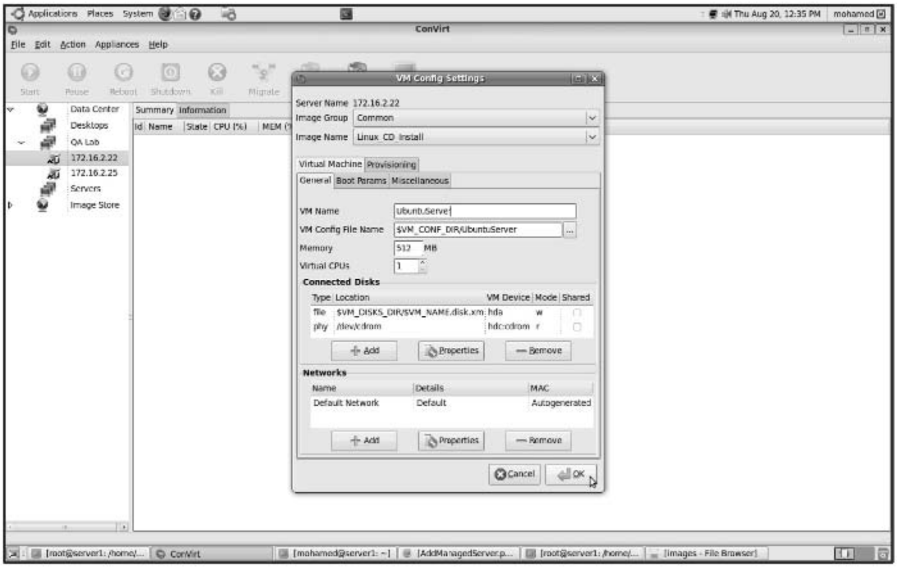

FIGURE 5.13. Configuring virtual machine.

the "provisioning" tab, and changing the VM_DISKS_DIR to point to the location of your shared NFS.

- Start your VM (Figures 5.14 and 5.15), and make sure the installation media of the operating system you need is placed in drive, in order to use it for booting the new VM and proceed in the installation process; then start the installation process as shown in Figure 5.16.
- Once the installation finishes, you can access your provisioned virtual machine from the consol icon on the top of your ConVirt management console.
- Reaching this step, you have created your first managed server and provisioned virtual machine. You can repeat the same procedure to add the second managed server in your pool to be ready for the next step of migrating one virtual machine from one server to the other.

5.5.1 VM Life Cycle and VM Monitoring

You can notice through working with ConVirt that you are able to manage the whole life cycle of the virtual machine; start, stop, reboot, migrate, clone, and so on. Also, you noticed how easy it is to monitor the resources of the managed server and to monitor the virtual machine's guests that help you balance and control the load on these managed servers once needed. In the next section, we are going to discuss how easy it is to migrate a virtual machine from host to host.

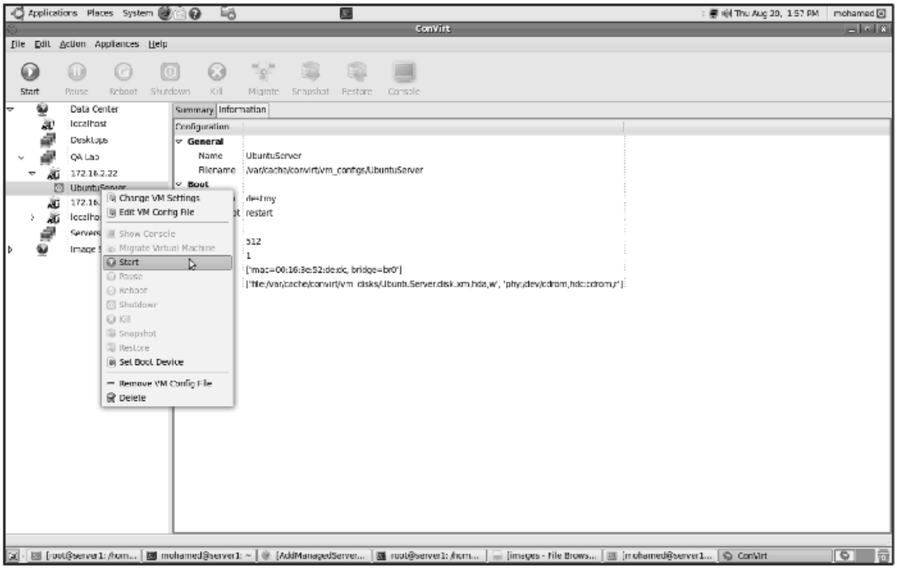

FIGURE 5.14. Provisioned VM ready to be started.

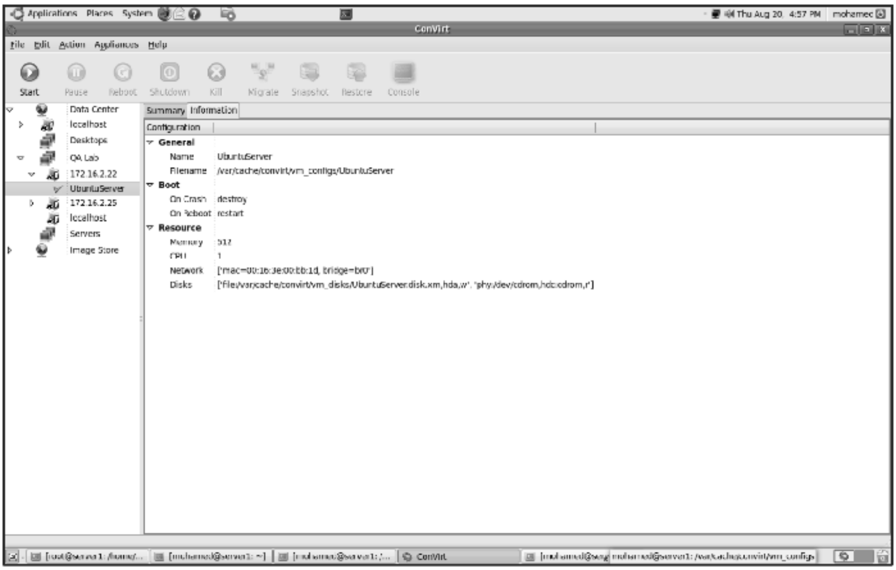

FIGURE 5.15. Provisioned VM started.

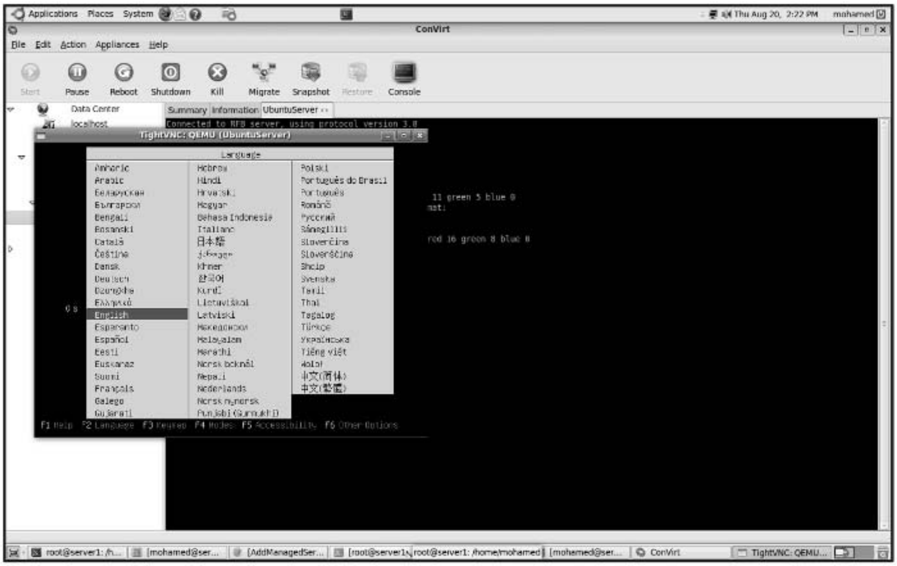

FIGURE 5.16. VM booting from the installation CD to start the installation process.

5.5.2 Live Migration

ConVirt tool allows running virtual machines to be migrated from one server to another [29].This feature makes it possible to organize the virtual machine to physical machine relationship to balance the workload; for example, a VM needing more CPU can be moved to a machine having available CPU cycles, or, in other cases, like taking the host machine for maintenance. For proper VM migration the following points must be considered [29]:

- Shared storage for all Guest OS disks (e.g., NFS, or iSCSI).
- Identical mount points on all servers (hosts).
- The kernel and ramdisk when using para-virtualized virtual machines should, also, be shared. (This is not required, if pygrub is used.)
- Centrally accessible installation media (iso).
- It is preferable to use identical machines with the same version of virtualization platform.
- Migration needs to be done within the same subnet.

Migration Process in ConVirt

- To start the migration of a virtual machine from one host to the other, select it and choose a migrating virtual machine, as shown in Figure 5.17.
- You will have a window containing all the managed servers in your data center (as shown in Figure 5.18). Choose one as a destination and start

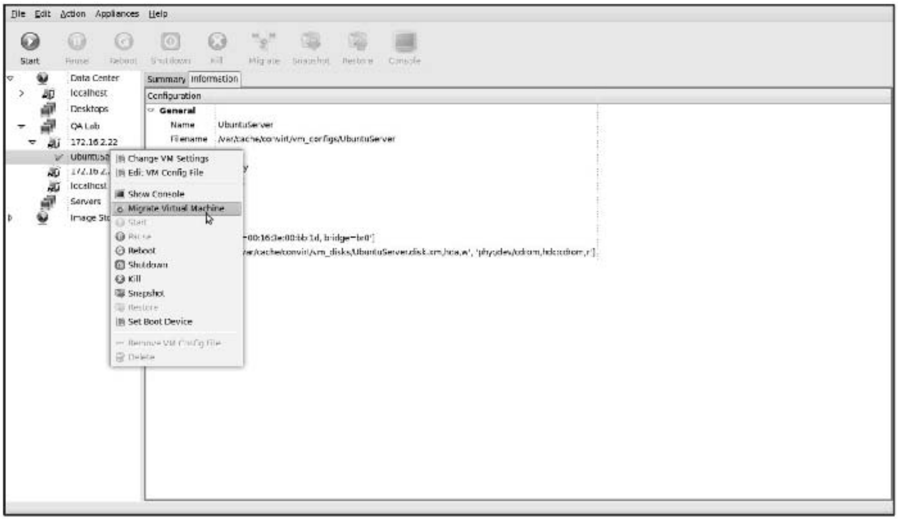

FIGURE 5.17. VM migration.

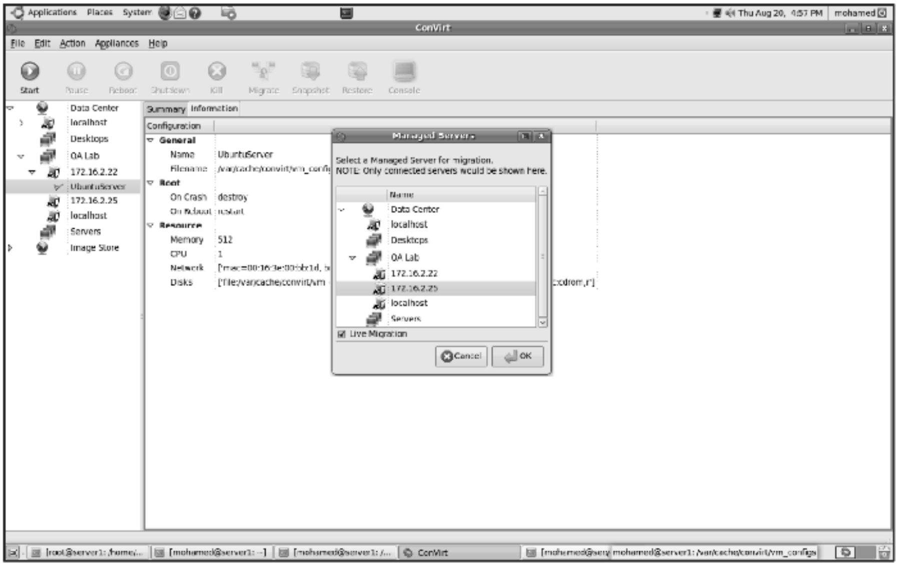

FIGURE 5.18. Select the destination managed server candidate for migration.

migration, or drag the VM and drop it on to another managed server to initiate migration.

- Once the virtual machine has been successfully placed and migrated to the destination host, you can see it still living and working (as shown in Figure 5.19).

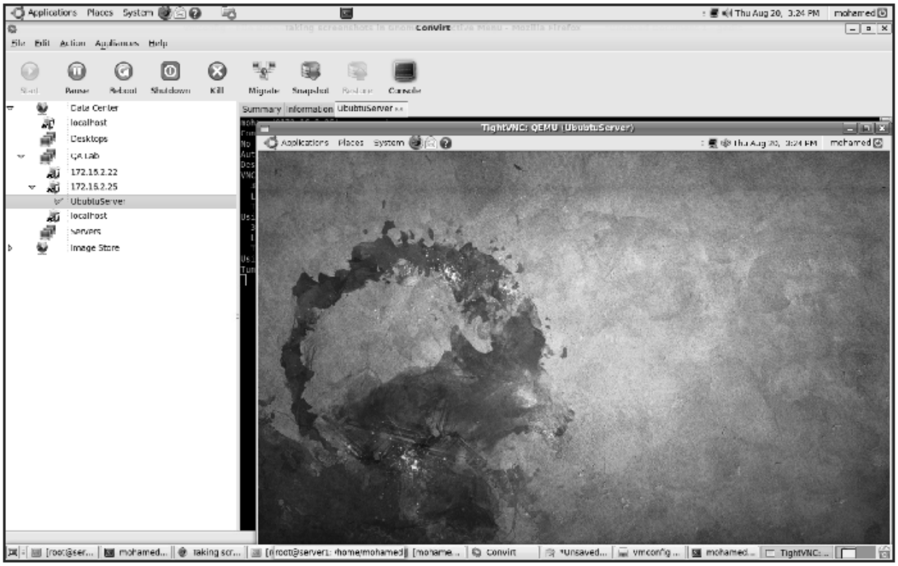

FIGURE 5.19. VM started on the destination server after migration.

5.5.3 Final Thoughts about the Example

This is just a demonstrating example of how to provision and migrate virtual machines; however, there are more tools and vendors that offer virtual infrastructure's management like Citrix XenServer, VMware vSphere, and so on.

5.6 PROVISIONING IN THE CLOUD CONTEXT

In the cloud context, we shall discuss systems that provide the virtual machine provisioning and migration services; Amazon EC2 is a widely known example for vendors that provide public cloud services. Also, Eucalyptus and Open-Nebula are two complementary and enabling technologies for open source cloud tools, which play an invaluable role in infrastructure as a service and in building private, public, and hybrid cloud architecture.

Eucalyptus is a system for implementing on-premise private and hybrid clouds using the hardware and software's infrastructure, which is in place without modification. The current interface to Eucalyptus is compatible with Amazon's EC2, S3, and EBS interfaces, but the infrastructure is designed to support multiple client-side interfaces. Eucalyptus is implemented using commonly available Linux tools and basic Web service's technologies [30]. Eucalyptus adds capabilities, such as end-user customization, self-service provisioning, and legacy application support to data center's virtualization's features, making the IT customer's service easier [11]. On the other hand, OpenNebula is a virtual infrastructure manager that orchestrates storage, network, and virtualization technologies to enable the dynamic placement of multi-tier services on distributed infrastructures, combining both data center's resources and remote cloud's resources according to allocation's policies. OpenNebula provides internal cloud administration and user's interfaces for the full management of the cloud's platform.

5.6.1 Amazon Elastic Compute Cloud

The Amazon EC2 (Elastic Compute Cloud) is a Web service that allows users to provision new machines into Amazon's virtualized infrastructure in a matter of minutes; using a publicly available API (application programming interface), it reduces the time required to obtain and boot a new server. Users get full root access and can install almost any OS or application in their AMIs (Amazon Machine Images). Web services APIs allow users to reboot their instances remotely, scale capacity quickly, and use on-demand service when needed; by adding tens, or even hundreds, of machines. It is very important to mention that there is no up-front hardware setup and there are no installation costs, because Amazon charges only for the capacity you actually use.

EC2 instance is typically a virtual machine with a certain amount of RAM, CPU, and storage capacity.

Setting up an EC2 instance is quite easy. Once you create your AWS (Amazon Web service) account, you can use the on-line AWS console, or simply download the offline command line's tools to start provisioning your instances.

Amazon EC2 provides its customers with three flexible purchasing models to make it easy for the cost optimization:

- ***On-Demand*** instances, which allow you to pay a fixed rate by the hour with no commitment.
- ***Reserved*** instances, which allow you to pay a low, one-time fee and in turn receive a significant discount on the hourly usage charge for that instance. It ensures that any reserved instance you launch is guaranteed to succeed (provided that you have booked them in advance). This means that users of these instances should not be affected by any transient limitations in EC2 capacity.
- ***Spot*** instances, which enable you to bid whatever price you want for instance capacity, providing for even greater savings, if your applications have flexible start and end times.

Amazon and Provisioning Services. Amazon provides an excellent set of tools that help in provisioning service; Amazon Auto Scaling [30] is a set of command line tools that allows scaling Amazon EC2 capacity up or down automatically and according to the conditions the end user defines. This feature ensures that the number of Amazon EC2 instances can scale up seamlessly during demand spikes to maintain performance and can scale down automatically when loads diminish and become less intensive to minimize the costs. Auto Scaling service and CloudWatch [31] (a monitoring service for AWS cloud resources and their utilization) help in exposing functionalities required for provisioning application services on Amazon EC2.

Amazon Elastic Load Balancer [32] is another service that helps in building fault-tolerant applications by automatically provisioning incoming application workload across available Amazon EC2 instances and in multiple *availability zones*.

5.6.2 Infrastructure Enabling Technology

Offering infrastructure as a service requires software and platforms that can manage the Infrastructure that is being shared and dynamically provisioned. For this, there are three noteworthy technologies to be considered: Eucalyptus, OpenNebula, and Aneka.

5.6.3 Eucalyptus

Eucalyptus [11] is an open-source infrastructure for the implementation of cloud computing on computer clusters. It is considered one of the earliest tools developed as a surge computing (in which data center's private cloud could augment its ability to handle workload's spikes by a design that allows it to send overflow work to a public cloud) tool. Its name is an acronym for "elastic utility computing architecture for linking your programs to useful systems." Here are some of the Eucalyptus features [11]:

- Interface compatibility with EC2, and S3 (both Web service and Query/REST interfaces).
- Simple installation and deployment.
- Support for most Linux distributions (source and binary packages).
- Support for running VMs that run atop the Xen hypervisor or KVM. Support for other kinds of VMs, such as VMware, is targeted for future releases.
- Secure internal communication using SOAP with WS security.
- Cloud administrator's tool for system's management and user's accounting.
- The ability to configure multiple clusters each with private internal network addresses into a single cloud.

Eucalyptus aims at fostering the research in models for service's provisioning, scheduling, SLA formulation, and hypervisors' portability.

Eucalyptus Architecture. Eucalyptus architecture, as illustrated in Figure 5.20, constitutes each high-level system's component as a stand-alone Web service with the following high-level components [11].

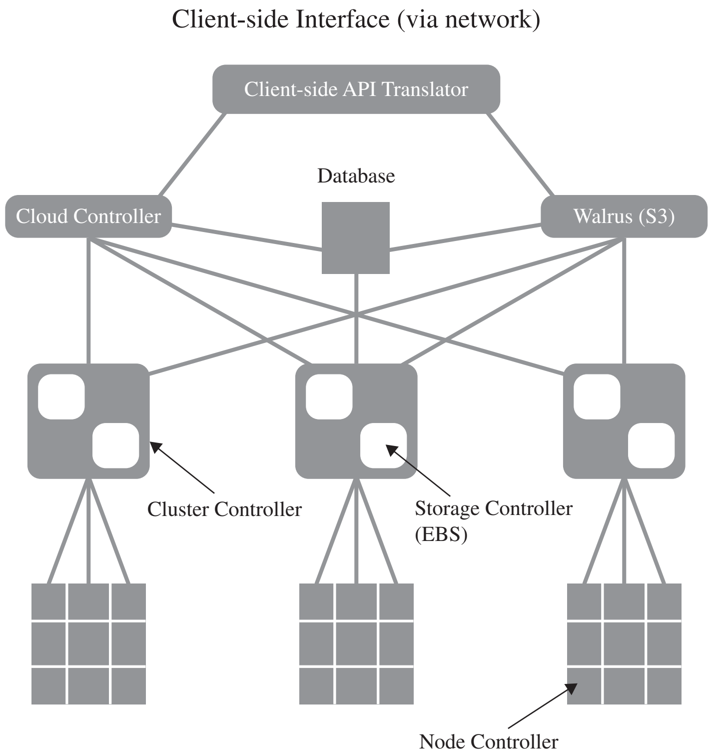

FIGURE 5.20. Eucalyptus high-level architecture.

- **Node controller (NC)** controls the execution, inspection, and termination of VM instances on the host where it runs.
- **Cluster controller (CC)** gathers information about and schedules VM execution on specific node controllers, as well as manages virtual instance network.
- **Storage controller (SC)** is a put/get storage service that implements Amazon's S3 interface and provides a way for storing and accessing VM images and user data.
- **Cloud controller (CLC)** is the entry point into the cloud for users and administrators. It queries node managers for information about resources, makes high-level scheduling decisions, and implements them by making requests to cluster controllers.
- **Walrus (W)** is the controller component that manages access to the storage services within Eucalyptus. Requests are communicated to Walrus using the SOAP or REST-based interface.

Its design is an open and elegant one. It can be very beneficial in testing and debugging purposes before deploying it on a real cloud. For more details about Eucalyptus architecture and design, check reference 11.

Ubuntu Enterprise Cloud and Eucalyptus. Ubuntu Enterprise Cloud (UEC) [33] is a new initiative by Ubuntu to make it easier to provision, deploy, configure, and use cloud infrastructures based on Eucalyptus. UEC brings Amazon EC2-like infrastructure's capabilities inside the firewall.

This is by far the simplest way to install and try Eucalyptus. Just download the Ubuntu server version and install it wherever you want. UEC is also the first open source project that lets you create cloud services in your local environment easily and leverage the power of cloud computing.

5.6.4 VM Dynamic Management Using OpenNebula

OpenNebula [12] is an open and flexible tool that fits into existing data center's environments to build any type of cloud deployment. OpenNebula can be primarily used as a virtualization tool to manage your virtual infrastructure, which is usually referred to as private cloud. OpenNebula supports a hybrid cloud to combine local infrastructure with public cloud-based infrastructure, enabling highly scalable hosting environments. OpenNebula also supports public clouds by providing cloud's interfaces to expose its functionality for virtual machine, storage, and network management. OpenNebula is one of the technologies being enhanced in the Reservoir Project [14], European research initiatives in virtualized infrastructures, and cloud computing.

OpenNebula architecture is shown in Figure 5.21, which illustrates the existence of public and private clouds and also the resources being managed by its virtual manager.

OpenNebula is an open-source alternative to these commercial tools for the dynamic management of VMs on distributed resources. This tool is supporting several research lines in advance reservation of capacity, probabilistic admission control, placement optimization, resource models for the efficient management of groups of virtual machines, elasticity support, and so on. These research lines address the requirements from both types of clouds namely, private and public.

OpenNebula and Haizea. Haizea is an open-source virtual machine-based lease management architecture developed by Sotomayor et al. [34]; it can be used as a scheduling backend for OpenNebula. Haizea uses leases as a fundamental *resource provisioning* abstraction and implements those leases as virtual machines, taking into account the overhead of using virtual machines when scheduling leases. Haizea also provides advanced functionality such as [35]:

- Advance reservation of capacity.
- Best-effort scheduling with backfilling.

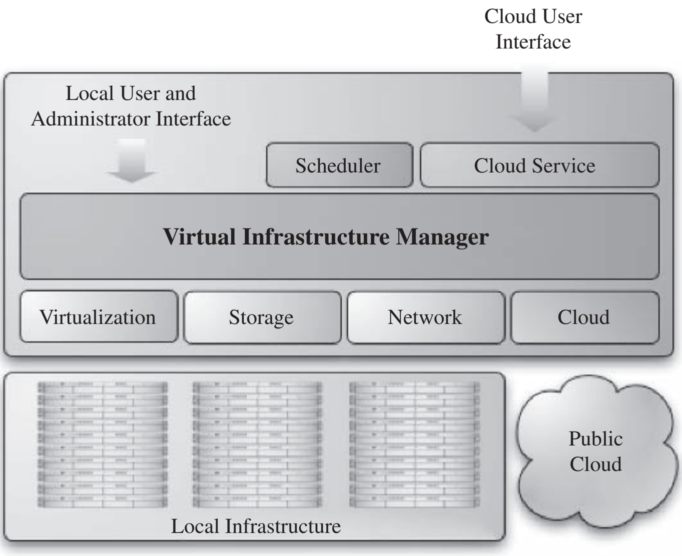

FIGURE 5.21. OpenNebula high-level architecture [14].

- Resource preemption (using VM suspend/resume/migrate).
- Policy engine, allowing developers to write pluggable scheduling policies in Python.

5.6.5 Aneka

Manjrasoft Aneka [10] is a .NET-based platform and framework designed for building and deploying distributed applications on clouds. It provides a set of APIs for transparently exploiting distributed resources and expressing the business logic of applications by using the preferred programming abstractions. Aneka is also a market-oriented cloud platform since it allows users to build and schedule applications, provision resources, and monitor results using pricing, accounting, and QoS/SLA services in private and/or public cloud environments.

It allows end users to build an enterprise/private cloud setup by exploiting the power of computing resources in the enterprise data centers, public clouds such as Amazon EC2 [4], and hybrid clouds by combining enterprise private clouds managed by Aneka with resources from Amazon EC2 or other enterprise clouds built and managed using technologies such as XenServer.

Aneka also provides support for deploying and managing clouds. By using its *Management Studio* and a set of Web interfaces, it is possible to set up either public or private clouds, monitor their status, update their configuration, and perform the basic management operations.

Aneka Architecture. Aneka platform architecture [10], as illustrated in Figure 5.22, consists of a collection of physical and virtualized resources connected through a network. Each of these resources hosts an instance of the Aneka container representing the runtime environment where the distributed applications are executed. The container provides the basic management features of the single node and leverages all the other operations on the services that it is hosting. The services are broken up into fabric, foundation, and execution services. Fabric services directly interact with the node through the platform abstraction layer (PAL) and perform hardware profiling and *dynamic resource provisioning*. Foundation services identify the core system of the Aneka middleware, providing a set of basic features to enable Aneka containers to perform specialized and specific sets of tasks. Execution services directly deal with the scheduling and execution of applications in the cloud.

5.7 FUTURE RESEARCH DIRECTIONS

Virtual machine provision and migration services take their place in research to achieve the best out of its objectives, and here is a list of potential areas' candidates for research:

- Self-adaptive and dynamic data center.

Data centers exist in the premises of any hosting or ISPs that host different Web sites and applications. These sites are being accessed at different timing pattern (morning hours, afternoon, etc.). Thus, workloads against these sites need to be tracked because they vary dynamically over time. The sizing of host machines (the number of virtual machines that host these applications) represents a challenge, and there is a potential research area over here to study the performance impact and overhead due to this dynamic creation of virtual machines hosted in these self-adaptive data centers, in order to manage Web sites properly.

Study of the performance in this dynamic environment will also tackle the the balance that should be exist between a rapid response time of individual applications, the overall performance of the data, and the high availability of the applications and its services.

- Performance evaluation and workload characterization of virtual workloads.

It is very invaluable in any virtualized infrastructure to have a notion about the workload provisioned in each VM, the performance's impacts due to the hypervisors layer, and the overhead due to consolidated workloads for such systems; but yet, this is not a deterministic process. Single-workload benchmark is useful in quantifying the virtualization overhead within a single VM, but not useful in a whole virtualized environment with multiple isolated VMs with varying workloads on each, leading to the inability of capturing the system's

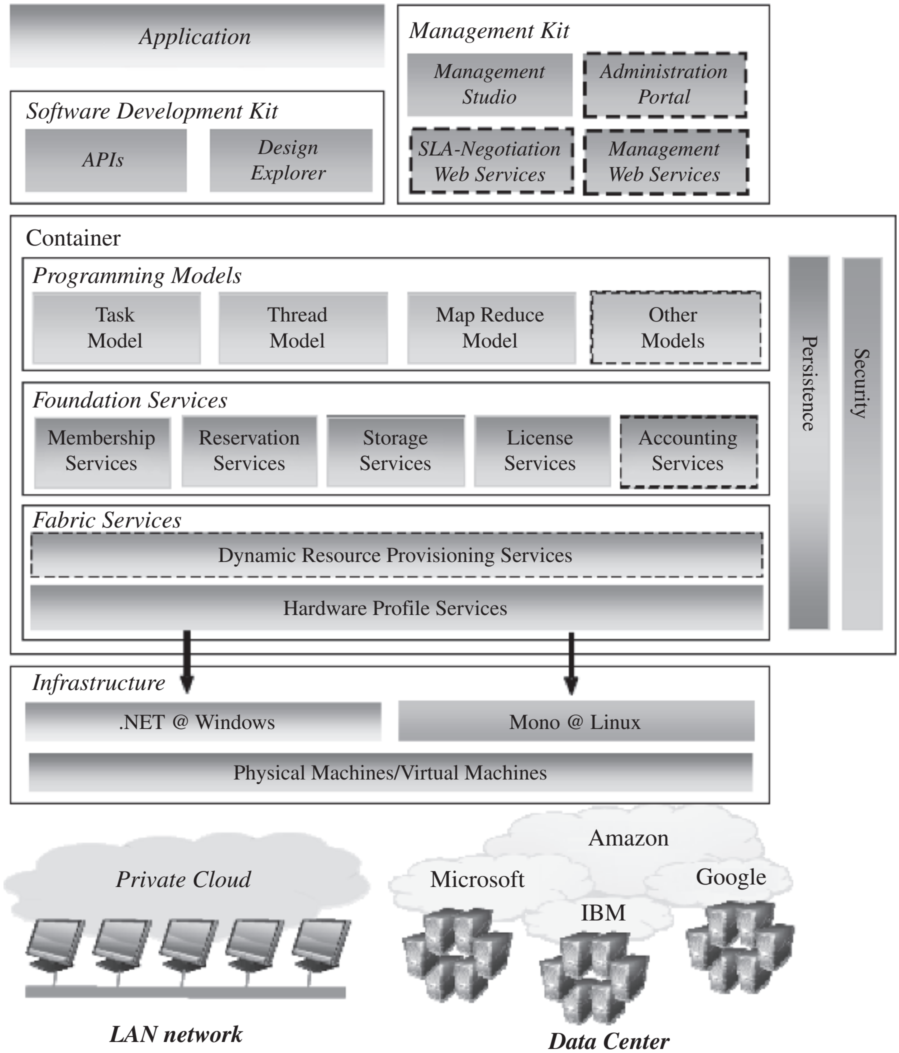

FIGURE 5.22. Manjras oft Aneka layered architecture [10].

behavior. So, there is a big need for a common workload model and methodology for virtualized systems; thus benchmark's results can be compared across different platforms. It will help in the dynamic workload's relocation and migrations' services.

- One of the potential areas that worth study and investigation is the development of fundamental tools and techniques that facilitate the integration and provisioning of distributed and hybrid clouds in federated way, which is critical for enabling of composition and deployment of elastic application services [35, 36].

- High-performance data scaling in private and public cloud environments.

Organizations and enterprises that adopt the cloud computing architectures can face lots of challenges related to (a) the elastic provisioning of compute clouds on their existing data center's infrastructure and (b) the inability of the data layer to scale at the same rate as the compute layer. So, there is a persisting need for implementing systems that are capable of scaling data with the same pace as scaling the infrastructure, or to integrate current infrastructure elastic provisioning systems with existing systems that are designed to scale out the applications and data layers.

- Performance and high availability in clustered VMs through live migration.

Clusters are very common in research centers, enterprises, and accordingly in the cloud. For these clusters to work in a proper way, there are two aspects of great importance, namely, *high availability,* and *high performance* service. This can be achieved through clusters of virtual machines in which high available applications can be achieved through the live migration of the virtual machine to different locations in the cluster or in the cloud. So, the need exists to (a) study the performance, (b) study the performance's improvement opportunities with regard to the migrations of these virtual machines, and (c) decide to which location the machine should be migrated.

- VM scheduling algorithms.
- Accelerating VMs live migration time.
- Cloud-wide VM migration and memory de-duplication.

Normal VM migration is being done within the same physical site location (campus, data center, lab, etc.). However, migrating virtual machines between different locations is an invaluable feature to be added to any virtualization management's tools. For more details on memory status, storage relocation, and so on; check the patent pending technology about this topic [37]. Considering such setup can enable faster and longer-distance VM migrations, cross-site load balancing, power management, and de-duplicating memory throughout multiple sites. It is a rich area for research.

- Live migration security.

Live migration security is a very important area of research, because several security's vulnerabilities exist; check reference 38 for an empirical exploitation of live migration.

- Extend migration algorithm to allow for priorities.
- Cisco initiative UCS (Unified Commuting System) and its role in dynamic just-in-time provisioning of virtual machines and increase of business agility [39].

5.8 CONCLUSION

Virtual machines' provisioning and migration are very critical tasks in today's virtualized systems, data center's technology, and accordingly the cloud computing services.

They have a huge impact on the continuity, and availability of business. In a few minutes, you can provision a complete server with all its appliances to perform a particular functionality, or to offer a service. In a few milliseconds, you can migrate a virtual machine hosted on a physical server within a clustered environment to a completely different server for the purpose of maintenance, workloads' needs, and so on. In this chapter, we covered VM provisioning and migration services techniques, as well as tools and concepts, and also shed some light on potential areas for research.

REFERENCES

1. D. Chisnall, *The Definitive Guide to the Xen Hypervisor*, Upper Saddle River, NJ, Prentice Hall, 2008.
2. M. El-Refaey and M. Rizkaa, Virtual systems workload characterization: An overview, in *Proceedings of the 18th IEEE International Workshops on Enabling Technologies: Infrastructures for Collaborative Enterprises, WETICE 2009*, Groningen, The Netherlands, 29 June–1 July 2009.
3. A. T. Velte, T. J. Velte, and R. Elsenpeter, *Cloud Computing: A Practical Approach*, McGraw-Hill, New York, 2010.
4. Amazon Elastic Compute Cloud (Amazon EC2), http://aws.amazon.com/ec2/, March 15, 2010.
5. Cloud Hosting, Cloud Computing, Hybrid Infrastructure from GoGrid, http://www.gogrid.com/, March 19, 2010.
6. JoyentCloud Computing Companies: Domain, Application & Web Hosting Services, http://www.joyent.com/, March 10, 2010.
7. Rackspace hosting, http://www.rackspace.com/index.php, March 10, 2010.
8. AppNexus—Home, http://www.appnexus.com/, March 9, 2010.
9. FlexiScale cloud computing and hosting: instant Windows and Linux cloud servers on demand, http://www.flexiscale.com/, March 12, 2010.
10. C. Vecchiola, X. Chu, and R. Buyya, Aneka: A Software Platform for .NET-based cloud computing, high speed and large scale scientific computing, in Advances in Parallel Computing, W. Gentzsch, L. Grandinetti, G. Joubert (eds.), ISBN: 978-1-60750-073-5, IOS Press, Amsterdam, Netherlands, 2009, pp. 267–295.